

Master's Thesis Progress Documentation

Phase 4: Machine Learning Model Development

1. Thesis Title

Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

2. Purpose of This Document

This document summarizes the work completed during Phase 4 of the master's thesis.

The main purpose of this phase was to develop, evaluate, and compare several machine learning classification models for predicting customer churn.

The processed datasets created during Phase 3 were used as the input for model development. Different classification models were trained and compared using suitable evaluation metrics.

The final model was not selected only based on accuracy. Special attention was given to the identification of customers who were likely to churn.

This was important because the dataset contained fewer churn customers than non-churn customers. Therefore, metrics such as recall, precision, F1-score, ROC-AUC, and Precision-Recall AUC were also considered.

At the end of this phase, a Gradient Boosting model was selected. The final model generated customer-level churn predictions and churn probabilities.

These outputs will be used during the next phase for Explainable Artificial Intelligence analysis using SHAP.

3. Current Stage of the Thesis

The thesis has now completed the following stages:

- Research foundation and planning
- Dataset selection
- Exploratory Data Analysis
- Data cleaning
- Data preprocessing
- Machine learning model development

Phase 4 focused on creating the predictive component of the proposed visual analytics system.

The machine learning model is responsible for estimating the probability that an individual customer may leave the telecommunications company.

The model results will later be connected to SHAP explanation values and displayed through an interactive Power BI dashboard.

The complete project pipeline is currently:

Customer Data

- Data Preprocessing
 - Machine Learning Model
 - Churn Predictions and Probabilities
 - SHAP Explanations
 - Power BI Dashboard
 - User Evaluation
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4. Work Completed in Phase 4

4.1 Creation of the Model Development Notebook

A new Google Colab notebook was created for the machine learning model development phase.

The notebook was named:

03_model_training.ipynb

The notebook was designed to run independently using the saved outputs from Phase 3.

All model training, evaluation, comparison, tuning, threshold analysis, prediction generation, and output creation were completed inside this notebook.

4.2 Loading the Phase 3 Outputs

The following processed files from Phase 3 were loaded:

- X_train_processed.csv
- X_test_processed.csv
- y_train.csv
- y_test.csv
- train_reference.csv
- test_reference.csv

- feature_name_mapping.csv
- preprocessing_metadata.json

The model-development datasets contained:

- 5,625 training records
- 1,407 testing records
- 45 processed numerical features

The training dataset contained:

- 1,495 churn customers
- 4,130 non-churn customers

The testing dataset contained:

- 374 churn customers
- 1,033 non-churn customers

The positive prediction class was defined as:

- Churn = 1
- Non-churn = 0

4.3 Input Data Validation

Before training the models, several checks were performed to confirm that the Phase 3 outputs were correct.

The validation process confirmed that:

- The number of training feature rows matched the number of training target values.
- The number of testing feature rows matched the number of testing target values.
- The training and testing datasets contained the same 45 feature columns.
- The training data contained no missing values.
- The testing data contained no missing values.
- No infinite values were present.
- All processed features were numerical.
- The target variables contained only the values 0 and 1.
- The training customer reference records remained aligned with the training targets.
- The testing customer reference records remained aligned with the testing targets.

These checks confirmed that the processed data was suitable for machine learning.

4.4 Class Imbalance Consideration

The dataset contained more non-churn customers than churn customers.

The complete cleaned dataset contained:

- 5,163 non-churn customers
- 1,869 churn customers

This means that approximately 26.6% of the customers belonged to the churn class.

Because of this imbalance, accuracy was not used as the only model-selection criterion.

For example, a model could achieve relatively high accuracy by predicting most customers as non-churn. However, such a model may fail to identify customers who are actually likely to leave.

Therefore, the model evaluation focused on several metrics, particularly churn recall and the Precision-Recall AUC.

Oversampling methods such as SMOTE were not introduced because the initial models already achieved acceptable churn recall through class weighting and threshold adjustment.

4.5 Machine Learning Models Evaluated

Five classification models were initially evaluated.

Logistic Regression

Logistic Regression was used as a simple and interpretable baseline model.

It provided churn probability predictions and allowed comparison between a linear model and more complex tree-based methods.

Weighted Logistic Regression

A class-weighted version of Logistic Regression was also tested.

The purpose of class weighting was to give greater importance to the smaller churn class during model training.

This helped the model identify more actual churn customers.

Decision Tree

A Decision Tree model was trained to capture nonlinear relationships within the customer data.

The depth and minimum leaf size were controlled to reduce the risk of overfitting.

Random Forest

A Random Forest model was trained using a collection of decision trees.

This model was included to test whether an ensemble of trees could improve the prediction performance.

Gradient Boosting

A Gradient Boosting model was also trained.

Gradient Boosting develops several small decision trees in sequence. Each new tree attempts to improve the errors made by the previous trees.

This model was suitable for the structured Telco Churn dataset and can also support tree-based SHAP analysis during the next phase.

4.6 Cross-Validation Strategy

Five-fold stratified cross-validation was used to compare the models.

The training dataset was divided into five groups.

During each cross-validation round:

- Four groups were used to train the model.
- One group was used for validation.
- The process was repeated until every group had been used for validation.

Stratification ensured that each fold contained a similar proportion of churn and non-churn customers.

Cross-validation provided a more reliable evaluation than using only one training and validation split.

It also helped determine whether the models produced stable results across different parts of the training data.

The independent test dataset was not used during cross-validation.

4.7 Evaluation Metrics

The following metrics were used to evaluate each model.

Accuracy

Accuracy measures the percentage of all customers that were classified correctly.

Precision

Precision measures how many customers predicted as churn actually belonged to the churn class.

Higher precision means that fewer non-churn customers are incorrectly marked as churn.

Recall

Recall measures how many actual churn customers were correctly identified by the model.

Recall was especially important in this project because a false-negative prediction represents an actual churn customer who was missed by the model.

F1-score

The F1-score combines precision and recall into one measure.

It is useful when the dataset contains an imbalanced target variable.

ROC-AUC

ROC-AUC measures how well the model separates churn customers from non-churn customers across different probability thresholds.

Precision-Recall AUC

Precision-Recall AUC focuses more directly on performance for the positive churn class.

This metric was particularly useful because churn was the smaller class in the dataset.

Brier Score

The Brier Score measures the quality of the predicted probabilities.

A lower Brier Score indicates that the model probabilities are closer to the actual outcomes.

Confusion Matrix

The confusion matrix shows:

- True negatives
- False positives
- False negatives
- True positives

This made it possible to examine the types of correct and incorrect predictions made by the models.

4.8 Baseline Model Comparison

The five baseline models were compared using training cross-validation.

The main average results were:

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Gradient Boosting	0.8041	0.6719	0.5157	0.5832	0.8494	0.6683
Logistic Regression	0.8025	0.6538	0.5472	0.5955	0.8461	0.6601
Weighted Logistic Regression	0.7520	0.5218	0.8027	0.6324	0.8459	0.6589
Random Forest	0.7835	0.5777	0.6903	0.6289	0.8437	0.6539
Decision Tree	0.7358	0.5038	0.7967	0.6157	0.8301	0.6054

The Gradient Boosting model achieved the strongest baseline ROC-AUC and Precision-Recall AUC.

The standard Logistic Regression model produced a balanced result but identified only approximately 54.7% of actual churn customers.

Weighted Logistic Regression achieved a much higher churn recall of approximately 80.3%.

The Decision Tree also achieved high churn recall, but its overall ranking performance was lower.

The Random Forest produced acceptable validation results but showed a larger difference between training and validation performance.

4.9 Model Stability and Overfitting Analysis

The training results were compared with the validation results to examine overfitting.

Overfitting occurs when a model performs very well on the training data but does not perform similarly on unseen data.

The Gradient Boosting, Logistic Regression, Weighted Logistic Regression, and Decision Tree models showed relatively small training-validation differences.

The Random Forest showed a larger F1-score overfitting gap of approximately 0.1841.

This indicated that the Random Forest was learning the training data too closely.

Therefore, Random Forest was not selected as the final model.

The tuned Gradient Boosting model later achieved an F1-score overfitting gap of approximately 0.0165, indicating good stability.

4.10 Hyperparameter Tuning

Limited hyperparameter tuning was completed using only the training data.

The test dataset was not used during tuning.

Two strong model families were tuned:

- Logistic Regression
- Gradient Boosting

For Logistic Regression, different values of regularization strength and class weighting were evaluated.

For Gradient Boosting, the following settings were examined:

- Number of estimators
- Learning rate
- Tree depth
- Minimum number of samples per leaf
- Subsample size

The best Gradient Boosting settings were:

- Number of estimators: 100
- Learning rate: 0.05
- Maximum tree depth: 2
- Minimum samples per leaf: 30
- Subsample: 0.80

The shallow trees and low learning rate helped control model complexity.

4.11 Tuned Model Comparison

The two tuned candidate models produced the following cross-validation results:

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Tuned Gradient Boosting	0.8057	0.6837	0.5023	0.5786	0.8496	0.6693
Tuned Logistic Regression	0.8025	0.6538	0.5472	0.5955	0.8461	0.6601

The Tuned Gradient Boosting model achieved:

- The highest cross-validation Precision-Recall AUC
- The highest cross-validation ROC-AUC
- A small training-validation performance difference
- Stable results across the cross-validation folds
- Compatibility with tree-based SHAP explanations

The Tuned Logistic Regression model achieved a slightly higher F1-score at the standard threshold.

However, the Tuned Gradient Boosting model provided stronger probability-ranking performance.

Since the prediction threshold could later be adjusted using the training data, the Tuned Gradient Boosting model was selected as the final model.

4.12 Final Model Selection

The final selected model was:

Tuned Gradient Boosting Classifier

The selection was mainly based on:

- Cross-validation Precision-Recall AUC
- Cross-validation ROC-AUC
- Model stability
- Low overfitting risk
- Probability output availability
- Suitability for churn-risk ranking
- Compatibility with SHAP
- Suitability for customer-level prediction export
- Suitability for Power BI integration

The final model was trained using all 5,625 training records after the model-selection process was completed.

4.13 Prediction Threshold Analysis

The selected model generated a churn probability for every customer.

A probability threshold was required to convert the probability into a final churn or non-churn prediction.

The default threshold was:

0.50

This means:

- Probability of 0.50 or higher = predicted churn
- Probability below 0.50 = predicted non-churn

However, the default threshold identified only approximately half of the actual churn customers.

The purpose of the thesis model was not only to achieve high overall accuracy. It was also important to identify customers who may leave the company.

Therefore, different thresholds were evaluated using out-of-fold probabilities generated from the training dataset.

The test dataset was not used for threshold selection.

A target churn recall of approximately 80% was defined.

The selected threshold was:

0.27

This means:

- Probability of 0.27 or higher = predicted churn
- Probability below 0.27 = predicted non-churn

4.14 Training Threshold Comparison

Using the default threshold of 0.50, the out-of-fold training results were:

- Accuracy: 0.8057
- Precision: 0.6827
- Recall: 0.5023
- F1-score: 0.5788
- False positives: 349

- False negatives: 744

Using the selected threshold of 0.27, the results were:

- Accuracy: 0.7579
- Precision: 0.5293
- Recall: 0.8027
- F1-score: 0.6380
- False positives: 1,067
- False negatives: 295

Lowering the threshold produced the expected trade-off.

The model identified more actual churn customers, but it also incorrectly marked more non-churn customers as churn.

The selected threshold increased churn recall from approximately 50.2% to 80.3%.

It also improved the F1-score from approximately 57.9% to 63.8%.

The number of missed churn customers decreased from 744 to 295 in the out-of-fold training results.

4.15 Independent Test Evaluation

After model selection, tuning, and threshold selection were completed, the model was evaluated once using the independent test dataset.

The test dataset contained:

- 1,407 customers
- 374 actual churn customers
- 1,033 actual non-churn customers

At the default threshold of 0.50, the final test results were:

- Accuracy: 0.7960
- Precision: 0.6495
- Recall: 0.5053
- F1-score: 0.5684
- ROC-AUC: 0.8383
- Precision-Recall AUC: 0.6520
- Brier Score: 0.1386

At the selected threshold of 0.27, the final test results were:

- Accuracy: 0.7306

- Precision: 0.4959
- Recall: 0.8075
- F1-score: 0.6144
- ROC-AUC: 0.8383
- Precision-Recall AUC: 0.6520
- Brier Score: 0.1386

The selected threshold achieved the planned churn recall target.

It correctly identified approximately 80.75% of the actual churn customers in the independent test dataset.

4.16 Confusion Matrix Interpretation

At the selected threshold of 0.27, the final confusion matrix contained:

- True negatives: 726
- False positives: 307
- False negatives: 72
- True positives: 302

True Positives

The 302 true positives were customers who actually churned and were correctly predicted as churn customers.

These customers could potentially be selected for retention actions.

False Negatives

The 72 false negatives were customers who actually churned but were predicted as non-churn customers.

These are important errors because the company may fail to contact customers who are genuinely at risk of leaving.

False Positives

The 307 false positives were customers who did not churn but were predicted as churn customers.

These customers may unnecessarily receive retention communication or offers.

True Negatives

The 726 true negatives were non-churn customers who were correctly predicted as non-churn customers.

At the default threshold, the model missed 185 churn customers.

At the selected threshold, the number of missed churn customers decreased to 72.

Therefore, threshold adjustment reduced the number of false negatives by 113 customers.

This represents a reduction of approximately 61%.

4.17 Interpretation of the Final Model Performance

The final test accuracy of 73.06% was lower than the default-threshold accuracy of 79.60%.

However, this reduction was accepted because the selected threshold substantially improved churn detection.

The main business objective was to identify customers who may leave so that retention actions could be considered.

The final model identified approximately 81 out of every 100 actual churn customers.

The precision value of approximately 49.6% means that approximately half of the customers predicted as churn actually churned.

This reflects the trade-off created by using a lower and more sensitive prediction threshold.

The final ROC-AUC of 0.8383 indicated that the model had good ability to rank churn customers above non-churn customers.

The final Precision-Recall AUC of 0.6520 showed that the model performed meaningfully better than the basic churn rate of the dataset.

Overall, the selected model provided a useful balance between:

- Churn identification
 - Probability ranking
 - Model stability
 - Generalization
 - Explainability compatibility
 - Business usefulness
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4.18 Customer-Level Prediction Creation

Customer-level prediction files were created after the final test evaluation.

The model predictions were connected to the preserved customer reference records using customerID.

The main prediction output contained:

- customerID
- Actual churn status
- Predicted churn status
- Predicted churn label
- Default-threshold prediction
- Churn probability
- Selected prediction threshold

A second output file combined the predictions with the original customer information.

This file contains customer variables such as:

- Contract
- Tenure
- Monthly charges
- Total charges
- Internet service
- Payment method
- Technical support
- Online security
- Other customer characteristics
- Churn probability
- Final predicted churn status

These outputs will later support customer-level analysis in Power BI.

4.19 Training Out-of-Fold Prediction Creation

An out-of-fold training prediction file was also created.

Out-of-fold predictions were generated so that each training customer received a probability from a model that was not trained on that specific customer.

This approach reduces overly optimistic training predictions.

The file contains:

- Training row index
- customerID
- Actual churn status

- Out-of-fold churn probability
- Out-of-fold predicted churn
- Prediction threshold

This output can be used for additional analysis and comparison during later phases.

4.20 Prevention of Data Leakage

Several steps were applied to prevent data leakage.

The following rules were followed:

- The preprocessing pipeline from Phase 3 was fitted only on the training data.
- The test dataset was not used during cross-validation.
- The test dataset was not used during hyperparameter tuning.
- The test dataset was not used during model selection.
- The prediction threshold was selected using training out-of-fold probabilities.
- The test dataset was used only once for final independent evaluation.

These steps ensured that the final test results provided a fair estimate of model performance on unseen customers.

4.21 Saving the Selected Model

The final trained Gradient Boosting model was saved as:

selected_model.joblib

Saving the trained model makes it possible to load the exact same model during Phase 5.

The saved model was reloaded after creation.

The probabilities produced by the reloaded model were compared with the original model probabilities.

The comparison confirmed that the saved and reloaded models produced matching results.

This validated that the model had been stored correctly.

4.22 Saving the Phase 4 Outputs

All Phase 4 outputs were saved inside the following folder:

phase4_outputs

The following files were created:

- selected_model.joblib
- model_comparison_results.csv
- cross_validation_results.csv
- model_stability_results.csv
- tuned_model_comparison.csv
- threshold_analysis.csv
- oof_threshold_comparison.csv
- final_test_metrics.csv
- confusion_matrix_results.csv
- test_predictions.csv
- test_customer_predictions.csv
- training_oof_predictions.csv
- model_feature_names.csv
- best_model_parameters.json
- model_metadata.json

A ZIP archive was also created:

Phase_4_Model_Development_Outputs.zip

5. Purpose of the Main Output Files

selected_model.joblib

This file contains the final trained Tuned Gradient Boosting model.

It will be loaded during Phase 5 for SHAP explainability analysis.

model_comparison_results.csv

This file contains the cross-validation results for the five baseline classification models.

cross_validation_results.csv

This file contains the results from each individual cross-validation fold.

model_stability_results.csv

This file contains the training-validation differences used to examine model stability and overfitting.

tuned_model_comparison.csv

This file compares the tuned Logistic Regression and Tuned Gradient Boosting models.

threshold_analysis.csv

This file contains the precision, recall, F1-score, F2-score, accuracy, and predicted churn count for different prediction thresholds.

final_test_metrics.csv

This file contains the independent test results for both the default threshold and selected threshold.

confusion_matrix_results.csv

This file contains the true-negative, false-positive, false-negative, and true-positive results.

test_predictions.csv

This file contains customer IDs, actual churn values, model predictions, probabilities, and threshold information.

test_customer_predictions.csv

This file combines the original customer records with the final predictions and churn probabilities.

It will be useful during SHAP analysis and Power BI development.

training_oof_predictions.csv

This file contains leakage-safe out-of-fold predictions for the training customers.

model_feature_names.csv

This file records the order and names of the 45 processed model features.

best_model_parameters.json

This file records the selected model settings and selected probability threshold.

model_metadata.json

This file records the complete Phase 4 configuration, including:

- Selected model
- Feature names
- Training and testing record counts
- Cross-validation method

- Selected threshold
 - Final metrics
 - Python version
 - Scikit-learn version
 - Data-leakage prevention steps
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6. Tools and Libraries Used

The following tools and libraries were used during Phase 4:

- Google Colab
- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Joblib
- JSON
- ZIP file utilities

Scikit-learn was mainly used for:

- Logistic Regression
- Decision Tree classification
- Random Forest classification
- Gradient Boosting classification
- Stratified cross-validation
- Out-of-fold prediction generation
- Hyperparameter tuning
- Classification metrics
- ROC curve generation
- Precision-Recall curve generation
- Confusion matrix generation

Matplotlib was used to display:

- Confusion matrices
 - ROC curves
 - Precision-Recall curves
 - Threshold-performance trade-offs
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7. Key Model Development Decisions

The main decisions made during Phase 4 were:

- Use the exact processed datasets created in Phase 3.
 - Preserve the independent test dataset for final evaluation.
 - Evaluate several classification algorithms.
 - Use five-fold stratified cross-validation.
 - Consider the class imbalance during model evaluation.
 - Avoid selecting the model only according to accuracy.
 - Compare churn recall, precision, F1-score, ROC-AUC, and Precision-Recall AUC.
 - Examine training-validation differences for overfitting.
 - Perform limited hyperparameter tuning.
 - Select Tuned Gradient Boosting as the final model.
 - Use training out-of-fold probabilities for threshold analysis.
 - Select a threshold of 0.27 to achieve approximately 80% churn recall.
 - Preserve the default threshold results for comparison.
 - Save customer-level probabilities and predictions.
 - Save the trained model for SHAP analysis.
 - Connect predictions back to customer IDs.
 - Save all results and configuration information for reproducibility.
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8. Key Outcomes

At the end of Phase 4:

- The processed training and testing datasets were loaded successfully.
 - All model input validation checks passed.
 - Five classification models were evaluated.
 - Five-fold stratified cross-validation was completed.
 - Model stability and overfitting were examined.
 - Logistic Regression and Gradient Boosting were tuned.
 - Tuned Gradient Boosting was selected as the final model.
 - The final model achieved a test ROC-AUC of 0.8383.
 - The final model achieved a test Precision-Recall AUC of 0.6520.
 - A prediction threshold of 0.27 was selected.
 - The selected threshold achieved a churn recall of 0.8075.
 - The selected threshold achieved an F1-score of 0.6144.
 - The final model correctly identified 302 of the 374 churn customers.
 - The model missed 72 churn customers.
 - Customer-level prediction files were created.
 - The selected model was saved successfully.
 - The saved model was reloaded and validated.
 - All required Phase 4 outputs were created.
 - A ZIP archive containing the Phase 4 outputs was generated.
 - The model is now ready for SHAP explainability analysis.
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9. Connection to Explainable AI and Power BI

Phase 4 created the predictive engine of the proposed visual analytics system.

The model currently provides:

- Customer churn probability
- Predicted churn status
- Actual churn status
- Customer-level model output

However, the model does not yet provide a clear explanation of why an individual customer received a particular churn probability.

The next phase will use SHAP to calculate the contribution of each feature to the model prediction.

For example, SHAP may show that the following factors increased a customer's churn risk:

- Month-to-month contract
- High monthly charges
- Low tenure
- No technical support

It may also show factors that reduced the customer's churn risk.

The prediction and SHAP output files will later be imported into Power BI.

Power BI will be used to display:

- Customer churn probability
- Predicted churn status
- High-risk customer rankings
- Global feature importance
- Positive feature contributions
- Negative feature contributions
- Individual customer explanations
- Customer and segment comparisons
- Interactive filters
- Drill-through pages

This means that Python provides the prediction and explanation values, while Power BI provides the interactive visual interface.

10. Next Steps

The next phase of the thesis will focus on Explainable Artificial Intelligence using SHAP.

The planned tasks include:

1. Create a new notebook named 04_shap_analysis.ipynb.
2. Load the selected Gradient Boosting model.
3. Load the processed training and testing datasets.
4. Load the customer reference records.
5. Initialize the appropriate SHAP explainer.
6. Calculate SHAP values for the selected model.
7. Generate global feature-importance results.
8. Generate SHAP summary plots.
9. Examine how important features influence churn predictions.
10. Generate local explanations for individual customers.
11. Identify features that increase churn probability.
12. Identify features that reduce churn probability.
13. Connect SHAP values to customer IDs.
14. Create structured SHAP output files.
15. Prepare the prediction and explanation outputs for Power BI.
16. Save the SHAP analysis results.
17. Create the Phase 5 progress documentation.

11. Summary

Phase 4 focused on developing the machine learning prediction component of the thesis.

Five classification models were evaluated using five-fold stratified cross-validation.

The models included Logistic Regression, Weighted Logistic Regression, Decision Tree, Random Forest, and Gradient Boosting.

Model performance was evaluated using accuracy, precision, recall, F1-score, ROC-AUC, Precision-Recall AUC, confusion matrices, and probability-quality measures.

Model stability and overfitting were also examined.

Limited hyperparameter tuning was completed for Logistic Regression and Gradient Boosting.

The Tuned Gradient Boosting model was selected because it achieved the strongest cross-validation probability-ranking results while maintaining a low overfitting risk.

A prediction threshold of 0.27 was selected using training out-of-fold probabilities.

The threshold was lowered from the standard value of 0.50 because the default threshold missed too many actual churn customers.

Using the selected threshold, the final model achieved a churn recall of 0.8075 on the independent test dataset.

This means that the model correctly identified approximately 81% of the actual churn customers.

The final model, evaluation results, prediction probabilities, customer-level outputs, model settings, and metadata were saved successfully.